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Inference Service Provider Evaluation Framework

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Technical Due Diligence & Service Quality Assessment

Owner	Version	Date
AMD ISV Partnership Team	1.0	September 2026

Applies To
Inference API Providers — Commercial Partnership Evaluation

Purpose & Scope

This framework standardizes how the AMD ISV Partnership team evaluates third-party inference API providers before recommending them to AMD partners or entering commercial agreements. It is **provider-agnostic and model-agnostic**, covering the full range of technical, security, commercial, and operational dimensions that determine whether a provider is production-ready for enterprise ISV workloads.

Output

A weighted composite score (1–5), a decision tier classification (Recommend / Conditional / Defer / Reject), and a structured findings report with conditions and recommended next steps.

Timeline

Full evaluation: 3–5 business days. Fast-track (performance + cost only): 1 day with automated scripts. Security & commercial review may require 1–2 additional weeks for vendor responses.

Prerequisites

API access with sufficient credit (≥\$200 recommended), model IDs, pricing sheet, vendor contact for Q&A, and NDA in place before sharing evaluation findings.

Evaluation Categories & Weights

#	CATEGORY	WEIGHT	RATIONALE
1	Inference Performance & Latency	25%	TTFT and throughput are the primary reason partners choose or reject a provider; directly affects end-user experience
2	Reliability & Availability	20%	Downtime and error rates have immediate revenue impact on partners running production applications
3	API Completeness & Compatibility	15%	Determines re-engineering cost to adopt the provider; OpenAI compatibility is the de facto standard
4	Security & Compliance	15%	Non-negotiable for enterprise partners in regulated industries; SOC 2 and data handling are board-level concerns
5	Cost Structure & Commercial Terms	12%	Total cost of ownership must be predictable; pricing transparency and volume flexibility affect ISV margin
6	Output Quality & Safety	8%	Factual accuracy, safety alignment, and output consistency determine partner liability exposure
7	Support & Operations	5%	Reflects long-term partnership viability; poor support is a leading indicator of churn risk
Total		100%	

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Inference Performance & Latency

25%

KEY METRICS

- TTFT p50 / p95 / p99 at c=1, 5, 10, 25, 50
- Token generation speed (tok/s) p50 at each concurrency
- End-to-end latency p50 / p99
- Latency degradation ratio (p99 / p50) — tail behavior
- Throughput at saturation (max stable RPS)
- Cold-start penalty (first request after idle)
- Streaming SSE first-byte vs. batch latency delta

ISV TARGET THRESHOLDS

METRIC	TARGET	ACCEPTABLE
TTFT p50 @ c=1	< 300ms	< 600ms
TTFT p99 @ c=1	< 800ms	< 2,000ms
TTFT p50 @ c=25	< 800ms	< 1,500ms
Tok/s p50 @ c=1	> 80	> 30
p99/p50 ratio	< 2×	< 4×
Error rate	0%	< 0.5%

* Reasoning models: TTFT targets relax by ~2× due to thinking token overhead. Disclose to latency-sensitive partners.

SCORING RUBRIC

SCORE	CRITERIA
5	TTFT p50 < 300ms @ c=1; p99 < 800ms; tok/s > 80; p99/p50 < 2×; 0% errors across all concurrency levels
4	TTFT p50 < 600ms @ c=1; p99 < 2s; tok/s > 30; < 0.5% errors; graceful degradation under load
3	TTFT p50 < 1,000ms @ c=1; p99 spikes present but < 5s; tok/s > 15; < 1% errors; noticeable degradation at c=25+
2	TTFT p50 > 1,000ms or p99 > 5s; tok/s < 15; error rate 1–3%; significant degradation under moderate load
1	TTFT p50 > 2,000ms; p99 > 10s; frequent errors; unstable or unusable under concurrency

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Reliability & Availability

20%

KEY METRICS

- Observed availability over soak test (≥ 2h @ steady RPS)
- Error rate (4xx / 5xx) during soak
- Published SLA uptime percentage
- P99 latency drift over time (hourly buckets)
- Rate-limit behavior — graceful 429 vs. silent drop
- Recovery time after induced error
- Status page freshness and incident history (90 days)

ISV TARGET THRESHOLDS

METRIC	TARGET	ACCEPTABLE
Observed availability	≥ 99.9%	≥ 99.5%
Error rate (soak)	< 0.1%	< 0.5%
Published SLA	99.9%	99.5%
SLA with penalties	Required	Negotiable
Latency drift (hourly)	< 10%	< 25%
Incident resolution	< 1h P1	< 4h P1

SCORING RUBRIC

SCORE	CRITERIA
5	≥ 99.9% observed; < 0.1% errors; published SLA with financial penalties; status page live; < 1h P1 resolution; no significant incidents in 90 days
4	≥ 99.5% observed; < 0.5% errors; SLA published (penalties negotiable); status page exists; < 4h P1 resolution
3	≥ 99.0% observed; < 1% errors; verbal SLA commitment; some incident history but resolved; status page incomplete
2	98–99% observed; 1–3% errors; no formal SLA; rate limiting opaque; multiple recent incidents
1	< 98% availability; > 3% errors; no SLA; frequent unexplained failures; no status visibility

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API Completeness & Compatibility

15%

KEY METRICS

- OpenAI API compatibility (`/v1/chat/completions` schema)
- Streaming SSE support (`stream: true`)
- Tool / function calling support with JSON schema
- Structured output / JSON mode
- System prompt support
- Token usage reporting accuracy (prompt / completion / cached)
- Context window size (tokens)
- Model versioning / pinning support
- Batch API availability
- SDK / client library availability
- Documentation completeness and accuracy
- Sandbox / playground environment

COMPATIBILITY CHECKLIST

FEATURE	STATUS
Chat completions endpoint	Required
Streaming SSE	Required
Tool / function calling	Recommended
JSON mode / structured output	Recommended
Usage token reporting	Required
Model version pinning	Recommended
Prompt caching	Recommended
Batch API	Optional
Embeddings endpoint	Optional

SCORING RUBRIC

SCORE	CRITERIA
5	Full OpenAI drop-in compatible; streaming, tool calling, JSON mode, accurate token reporting, model pinning, SDK, and complete docs
4	Compatible for core use cases; streaming works; tool calling present but may have edge-case gaps; docs mostly complete
3	Chat completions and streaming work; tool calling unverified or partial; token reporting present; docs thin
2	Non-standard API requiring adapter layer; streaming unreliable; token reporting inaccurate; poor docs
1	Significant incompatibility; no streaming; missing core fields; undocumented; requires major integration effort

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Security & Compliance

15%

KEY METRICS

- Data privacy policy: no training on customer prompts/responses
- Data retention period and deletion policy
- Encryption in transit (TLS 1.2+) and at rest (AES-256)
- SSL certificate validity and standard CA chain
- SOC 2 Type II certification (or equivalent)
- ISO 27001 / CSA STAR certification
- GDPR / CCPA compliance declaration
- Model isolation (shared inference vs. dedicated)
- Audit logging and access controls
- Penetration testing cadence
- Incident notification SLA
- HIPAA / FedRAMP (for regulated verticals)

SCORING RUBRIC

SCORE	CRITERIA
5	SOC 2 Type II + ISO 27001; no training on customer data (contractual); TLS 1.3; standard CA; dedicated inference option; GDPR DPA available; annual pen test
4	SOC 2 Type II in progress or completed; no training on data (policy); TLS 1.2+; GDPR compliant; audit logs available
3	SOC 2 Type I or self-attestation; no training policy stated but not contractual; standard TLS; GDPR partially addressed
2	No certifications; limited privacy policy; SSL issues; shared-only inference; data handling unclear
1	No security documentation; may train on customer data; non-standard TLS; unsuitable for any enterprise use

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Cost Structure & Commercial Terms

12%

KEY METRICS

- Input / output token price vs. comparable providers (GPT-4o benchmark)
- Cached token pricing (prompt cache hit rate potential)
- Reasoning token billing transparency (included in output?)
- Volume discount tiers and thresholds
- Rate limit tiers (RPM / TPM) per plan
- Overage handling (hard cap vs. throttle vs. auto-upgrade)
- Billing granularity (per-token vs. per-request vs. per-minute)
- Free trial / sandbox credit availability
- Contract flexibility (monthly vs. annual; minimum commits)
- Refund / credit policy for downtime

SCORING RUBRIC

SCORE	CRITERIA
5	> 50% cheaper than GPT-4o equivalent; published volume tiers; transparent rate limits; month-to-month available; clear overage policy; SLA credits
4	20–50% cheaper; volume discounts negotiable; rate limits documented; billing transparent; reasonable contract terms
3	Competitive or at-par pricing; some volume flexibility; rate limits partially documented; standard contract
2	Above-market pricing; rate limits undocumented; opaque billing; annual commit required; no SLA credits
1	Significantly above market; hidden fees; no pricing transparency; punitive contract terms

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Output Quality & Safety

8%

KEY METRICS

- Factual accuracy (knowledge base Q&A — 10 questions)
- Instruction following (format, length, keyword compliance)
- Output consistency / semantic overlap (same prompt × 3 runs)
- Safety refusal rate (5 adversarial prompts)
- Refusal phrasing consistency (keyword-detectable)
- Hallucination rate on domain-specific tasks
- Reasoning quality (chain-of-thought coherence)
- Long-context coherence (if > 32K context window)

SCORING RUBRIC

SCORE	CRITERIA
5	≥ 90% factual accuracy; 5/5 safety refusals; semantic overlap > 0.75; strong instruction following; consistent refusal phrasing
4	80–90% factual; 4–5/5 safety; semantic overlap 0.60–0.75; good instruction following with minor gaps
3	70–80% factual; 3–4/5 safety; semantic overlap 0.40–0.60; instruction following adequate; phrasing varies
2	60–70% factual; ≤ 3/5 safety; semantic overlap < 0.40; frequent instruction failures
1	< 60% factual; < 2/5 safety refusals; unreliable output; not suitable for customer-facing use

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Support & Operations

5%

KEY METRICS

- Response time to technical support tickets (P1 / P2 / P3)
- Dedicated account / technical contact availability
- Status page with real-time incident tracking
- Proactive incident notification (email / webhook)
- Developer documentation completeness
- Changelog / release notes cadence
- Model deprecation policy and advance notice period
- Roadmap transparency and feature request process

SCORING RUBRIC

SCORE	CRITERIA
5	Dedicated TAM; < 1h P1 SLA; proactive notifications; live status page; full docs; 90-day deprecation notice
4	Named contact; < 4h P1; status page; good docs; reasonable deprecation notice
3	Shared support; < 24h P1; status page exists but lags; docs adequate; deprecation policy unclear
2	Email-only support; slow response; no status page; thin docs; no deprecation notice
1	No formal support; community-only; no docs; silent deprecations

Hard Dealbreakers — Automatic Disqualification

Any single item below results in a **Reject** recommendation regardless of scores in other categories. These are not negotiable.

🚫 Trains on customer prompts or responses

Provider retains prompt/response data and uses it for model training without explicit opt-out. Immediate data privacy and IP exposure risk for ISV customers. Must be prohibited contractually.

🚫 Observed error rate > 3% at baseline concurrency

Error rate above 3% at c=1 or c=5 during a clean load test indicates fundamental service stability issues. Not acceptable for any production ISV workload.

🚫 No data privacy or data retention policy

Absence of a written data handling policy makes GDPR / CCPA compliance impossible. Cannot be recommended to any customer in a regulated industry or with EU users.

🚫 Non-standard or self-signed TLS certificate

SSL certificate that fails standard OS / browser trust stores (e.g., Windows, macOS, common Linux distributions) without workarounds. Blocks enterprise integration and signals poor operational maturity.

🚫 Safety refusal rate < 2/5 on standard adversarial prompt set

Fewer than 2 out of 5 hardcoded adversarial prompts refused indicates a model safety alignment gap that creates direct liability for ISV customers building consumer-facing products.

🚫 No published rate limits with no commitment to disclose

Undisclosed and uncommitted rate limits mean partners cannot capacity plan. Silent throttling or invisible hard caps will cause unpredictable production outages.

🚫 No path to SOC 2 certification within 12 months

For enterprise AMD ISV partners in financial services, healthcare, or government, SOC 2 Type II is required by their own compliance teams. Providers without a credible certification timeline cannot be recommended to these segments.

Technical Due Diligence Checklist

API & Integration

- ☐ OpenAI-compatible
/v1/chat/completions endpoint
verified TEST
- ☐ Streaming SSE tested and producing
correct data: chunks TEST
- ☐ Tool / function calling tested with a
sample schema TEST
- ☐ Token usage fields (prompt_tokens ,
completion_tokens) verified accurate
TEST
- ☐ Prompt caching API confirmed and cache
hit behavior verified TEST
- ☐ Reasoning token billing structure
documented (if reasoning model) DOC
- ☐ Model IDs confirmed — stable vs. aliased
(e.g., "latest") DOC
- ☐ Context window size documented and
verified DOC
- ☐ SSL certificate validated from Linux,
Windows, and macOS TEST
- ☐ Rate limit headers (x-ratelimit-*)
present in responses TEST
- ☐ 429 response includes retry-after
header TEST
- ☐ Batch API availability confirmed DOC

Security & Compliance

- ☐ Data processing agreement (DPA)
reviewed and signed LEGAL
- ☐ No-training-on-customer-data clause
confirmed contractually LEGAL
- ☐ Data retention period and deletion
process documented DOC
- ☐ SOC 2 Type II report (or Type I +
roadmap) obtained AUDIT
- ☐ TLS version ≥ 1.2 confirmed; certificate
CA is standard TEST
- ☐ Encryption at rest policy documented
DOC
- ☐ GDPR compliance declaration / DPA
reviewed LEGAL
- ☐ Shared vs. dedicated inference
infrastructure clarified DOC
- ☐ Subprocessor list reviewed and accepted
LEGAL
- ☐ Penetration test report (last 12 months)
reviewed AUDIT
- ☐ Security incident notification timeframe
confirmed ($< 72h$) DOC
- ☐ HIPAA BAA availability confirmed (if
healthcare partners) LEGAL

Commercial & SLA

- ☐ Published SLA uptime % reviewed and compared to observed availability **DOC**

- ☐ SLA financial penalties / service credits confirmed in contract **LEGAL**

- ☐ Rate limits per tier documented in writing (RPM / TPM / concurrent) **DOC**

- ☐ Volume discount thresholds and rates documented **DOC**

- ☐ Overage policy documented (auto-upgrade vs. hard cap vs. throttle) **DOC**

- ☐ Contract term and termination clause reviewed **LEGAL**

- ☐ Pricing lock-in period confirmed **DOC**

- ☐ Reseller / OEM pricing terms confirmed for AMD partner program **LEGAL**

Operations & Reliability

- ☐ Status page URL confirmed and tested for accuracy **TEST**

- ☐ 90-day incident history reviewed **DOC**

- ☐ Support ticket SLA (P1 / P2 / P3) documented and tested **DOC**

- ☐ Dedicated technical account contact confirmed **DOC**

- ☐ Model deprecation policy confirmed (advance notice period) **DOC**

- ☐ Geographic region availability documented (US / EU / APAC) **DOC**

- ☐ Multi-region / failover capability confirmed **DOC**

- ☐ Changelog / release notes process reviewed **DOC**

Vendor Interview Question Bank

Technical Questions

T1 What are your published rate limits (RPM / TPM / concurrent requests) at each pricing tier, and can we get them in writing as part of the contract? **TECH**

T2 Are all models on shared or dedicated infrastructure? Is a dedicated deployment option available, and at what cost? **TECH**

T3 How do you handle prompt caching? What is the cache TTL, and how is cache hit rate surfaced in the API response? **TECH**

T4 For reasoning models, are reasoning tokens billed at output token rates? Are they included in `completion_tokens` or tracked separately? **TECH**

T5 What is your model versioning and pinning strategy? If we pin to a specific model version, how long is it guaranteed to remain available? **TECH**

T6 Have you tested and validated your SSL/TLS certificate chain against Windows, macOS, and standard Linux trust stores? Are there any known compatibility issues? **TECH**

T7 What is your p99 TTFT at 50 concurrent requests under normal operating conditions, and do you have benchmarks to share? **TECH**

T8 Do you support tool / function calling with OpenAI-compatible JSON schema? Are there known gaps vs. the OpenAI spec? **TECH**

T9 What is the geographic distribution of your inference infrastructure? Can traffic be pinned to a specific region for data residency requirements? **TECH**

T10 What is your capacity planning model? How do you handle traffic spikes and what is the auto-scaling response time? **TECH**

Security & Compliance Questions

S1 Do you train your models on customer prompts or API responses? Is this committed contractually as a prohibition? **SECURITY**

S2 What is your data retention period for API logs and prompt/response data? How do customers request deletion? **SECURITY**

S3 Do you have SOC 2 Type II certification? If not, what is your target date and who is your auditor? **SECURITY**

S4 Can you provide your most recent penetration test executive summary? How frequently do you conduct pen tests? **SECURITY**

S5 What is your security incident notification SLA? Have you had any data breaches or security incidents in the last 24 months? **SECURITY**

Business & Commercial Questions

B1 What financial penalties or service credits apply when you miss your published SLA? Is there a cap on credits? **BUSINESS**

B2 What volume discount thresholds are available, and can pricing be locked in for 12–24 months? **BUSINESS**

B3 Do you offer a reseller or OEM pricing structure for AMD partners building products on your API? **BUSINESS**

B4 What is your company's funding status and runway? Are there any planned ownership changes or acquisitions? **BUSINESS**

B5 What is your 12-month roadmap for new models, API features, and infrastructure? What is your release cadence? **BUSINESS**

Evaluation Scorecard Template

Complete one scorecard per provider. Scores are 1–5; weighted score = $\Sigma(\text{score} \times \text{weight})$. Final recommendation determined by weighted score AND dealbreaker check.

CATEGORY	WEIGHT	RAW SCORE (1–5)	WEIGHTED SCORE	KEY FINDING
1. Inference Performance & Latency	25%	___ / 5	___	
2. Reliability & Availability	20%	___ / 5	___	
3. API Completeness & Compatibility	15%	___ / 5	___	
4. Security & Compliance	15%	___ / 5	___	
5. Cost Structure & Commercial Terms	12%	___ / 5	___	
6. Output Quality & Safety	8%	___ / 5	___	
7. Support & Operations	5%	___ / 5	___	
Total Weighted Score	100%	—	___ / 5.0	
Dealbreakers Triggered	Yes / No — list:			
Decision Tier	Recommend / Conditional / Defer / Reject			

Decision Tiers

✓ Recommend

Threshold: Weighted score ≥ 4.0 AND zero dealbreakers AND no category below 3.0

- Provider meets or exceeds enterprise-grade standards across all dimensions
- Safe to recommend to AMD ISV partners for production workloads
- Include in AMD partner marketplace or preferred provider list
- Standard commercial agreement; proceed to procurement

⚠ Recommend with Conditions

Threshold: Weighted score 3.0–3.9 AND zero dealbreakers AND no more than one category below 2.5

- Provider is viable but has specific gaps that must be addressed
- Document each condition with a resolution timeline and owner
- Recommend only for specific use cases that avoid the gap areas
- Require written commitment to resolve conditions before expanding partnership
- Re-evaluate in 90 days after conditions are addressed

II Defer

Threshold: Weighted score 2.0–2.9 OR zero dealbreakers but two+ categories below 2.5

- Provider shows promise but is not mature enough for enterprise recommendation today
- Document specific gaps and re-engage when provider can demonstrate improvement
- May recommend for internal / sandbox use only — not for customer-facing AMD partner products
- Schedule re-evaluation in 6 months with defined criteria for advancement to Conditional

⊘ Reject

Threshold: Any dealbreaker triggered OR weighted score < 2.0

- Do not recommend under any circumstances given current state
- Document dealbreaker(s) and share feedback with provider if relationship permits
- Can re-open evaluation only if provider resolves all dealbreakers and submits evidence
- Minimum 6-month waiting period before re-evaluation

Automated Test Scripts Reference

SCRIPT	CATEGORY	WHAT IT MEASURES	RUNTIME
<code>perf_test.py</code>	Performance	TTFT p50/p95/p99, tok/s, error rate across concurrency levels (c=1,5,10,25,50); streaming SSE	~30 min
<code>accuracy_test.py</code>	Quality & Safety	Factuality (10 Q&A), instruction following (5 tasks), safety refusals (5 adversarial prompts), semantic consistency	~10 min
<code>cost_estimator.py</code>	Cost	Per-request and monthly cost vs. GPT-4o, GPT-4o-mini, Claude Sonnet/Haiku across 3 workload profiles	~5 min
<code>soak_test.py</code>	Reliability	Availability, error rate, hourly latency drift over 2–24h @ steady RPS	2–24h

GitHub Actions Integration: All scripts are designed to run as a parallel CI/CD workflow. The `MANGOBOOST_API_KEY` (or equivalent) is stored as a repository secret. The report job runs after all test jobs complete and aggregates results into a single HTML report. See [.github/workflows/mangoboost_eval.yml](#) as a reference implementation.